

Qualifying Exam in Microeconomics
Arizona State University
June 2020

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. A consumer has preferences on the consumption set $X = \mathbb{R}_+^2$ represented by

$$u(x_1, x_2) = \phi(x_1) + x_2,$$

where ϕ is a *continuous* and *strictly increasing* function. In what follows you may make use of the fact that if ϕ is strictly concave, then u is strictly quasi-concave. The price vector is $p = (1, 1)$ and the consumer's wealth is $w \geq 0$. The consumer's ordinary (Walrasian) demand correspondence is $d(w)$ (suppressing the fixed price vector).

- (a) Let $\phi(x_1) = \sqrt{x_1}$. Derive the demand correspondence, $d(w)$. *Carefully justify your derivation.* (Remember that the consumption set is $\mathbb{R}_+^2$ and that the price of each good is 1.)
- (b) Return to the general case in which ϕ is continuous and strictly increasing, but not necessarily concave (and so preferences are not necessarily convex).
 - i. Prove that the demand for good 1 is normal in this sense: if $w_1 > w_0$ and $x^0 \in d(w_0)$, then there is a point $x^1 \in d(w_1)$ with $x_1^1 \geq x_1^0$.
 - ii. Does the following stronger property hold: if $w_1 > w_0$ and $x^t \in d(w_t)$ for $t = 0, 1$, then $x_1^1 \geq x_1^0$? In words: if wealth *increases*, the demand for good 1 *cannot decrease*. If so prove it, if not, give a counter-example (the simpler the better).

2. Consider the following competitive insurance market with adverse selection. There is a large number of identical insurance firms and heterogeneous consumers who have private information about their probability $\theta \in [\underline{\theta}, \bar{\theta}] \subseteq [0, 1]$ of suffering a loss of size $\ell \in (0, w)$, where w is the initial wealth of the consumers, and where θ is distributed according to a cumulative distribution F that has a strictly positive density f . Insurance firms compete by offering full insurance (the policy coverage is the entire loss ℓ) at a price (premium) t . Consumers have a Bernoulli utility function $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ that is strictly increasing and strictly concave. If a consumer with type θ does not buy insurance, then the consumer's expected utility is $\theta u(w - \ell) + (1 - \theta)u(w)$.

- (a) Define what a competitive equilibrium is in this market.
- (b) Prove that an equilibrium exists in this market.
- (c) Assume that f is uniform on $[0, 1]$. How many equilibria can there be? Show that there is always one in which the market breaks down. Assume in addition that $u(c) = \log c$, $w = 1$, $\ell = 0.8$, and show that there is an equilibrium with trade.
- (d) Suppose now there are only two types $0 < \theta_l < \theta_h < 1$ and let $\lambda \in (0, 1)$ be the fraction of high types in the market. Assume that there are two identical firms that compete by offering a finite number of contracts (x, t) where x is coverage, and consumers observe the contracts offered and choose the one that suits them best (or none of them). Set up the equations that describe the contracts that will be accepted by the two types of consumers in any subgame perfect equilibrium (in pure strategies) in this market, and draw a picture depicting these contracts (with t on the vertical axis and x on the horizontal one). In less than ten lines, explain how you would complete the equilibrium analysis of the screening game.

3. The following exchange economy has two goods and two agents. Agents are indexed by $i = 1, 2$. Goods are indexed by $\ell = 1, 2$. For all i , preferences are given by the utility function utility function

$$u_i(x_{1i}, x_{2i}) = (x_{1i})^{1/2} + (x_{2i})^{1/2}$$

where for $x_{\ell i}$ denotes the amount of good ℓ consumed by agent i . Agent 1's endowments is $\omega_1 = (2, 0)$ and agent 2's endowment is $\omega_2 = (0, 2)$.

- (a) Find the excess demand function for each agent and the aggregate excess demand for the economy.
- (b) Use the excess demand functions to identify a Walrasian equilibrium price.
- (c) For an arbitrary excess demand function $z(p)$ defined for $p \in \mathbb{R}_{++}^2$, give conditions which taken together guarantee that a Walrasian equilibrium exists.
Does the aggregate excess demand function in part (a) satisfy the conditions you provided.
- (d) Does the exchange economy described below have a Walrasian equilibrium? If so identify one. If not, carefully explain why not.

$$u_1(x_{11}, x_{21}) = \max\{x_{11}, x_{21}\} \text{ and } \omega_1 = (1, 0)$$

$$u_2(x_{12}, x_{22}) = \min\{x_{12}, x_{22}\} \text{ and } \omega_2 = (0, 1)$$

4. An *exchange economy* with I consumers is a list $\mathcal{E} = \left(\{u_i\}_{i=1}^I, \{\omega_i\}_{i=1}^I \right)$, where, for consumer $i = 1, \dots, I$, the function $u_i : \mathbb{R}_+^L \rightarrow \mathbb{R}$ represents consumer i 's preferences on the consumption set $\mathbb{R}_+^L$; and $\omega_i \in \mathbb{R}_+^L$ is i 's *nonzero* endowment of goods. Prices are normalized to lie in $\Delta = \{p \in \mathbb{R}_+^L \mid \sum p_i = 1\}$, the nonnegative unit simplex.

Assume that, for each $i = 1, \dots, I$, u_i is *continuous* and *strictly increasing* ($x > y \Rightarrow u_i(y) > u_i(x)$), but that is all that is assumed about each u_i . Let's agree on this notation:

- $x = \sum_{i=1}^I x_i$ (the aggregate consumption plan);
- $\omega = \sum_{i=1}^I \omega_i$ (the aggregate endowment); and
- $z = x - \omega$ (the aggregate excess demand vector).

- (a) Define precisely the notion of a *competitive (or Walrasian) equilibrium* of $\mathcal{E}$.
- (b) Arrow-Debreu (1954) formulate an *abstract economy* that adds to the I consumers of $\mathcal{E}$ a $(I + 1)$ -st actor, an *auctioneer*. The auctioneer chooses a price p in Δ . Preferences of the auctioneer over prices and aggregate consumption plans are represented by $p \cdot z$ (recall that $z = x - \omega$ is aggregate excess demand). As usual, given $p \in \Delta$, each consumer i chooses a consumption plan x_i in i 's budget set,

$$B_i(p) = \{x_i \in \mathbb{R}_+^L \mid p \cdot x_i \leq p \cdot \omega_i\}.$$

An *equilibrium of the abstract economy* is a profile $(x_1^*, \dots, x_I^*, p^*)$ such that

- I. for each consumer i , x_i^* maximizes u_i on $B_i(p^*)$; and
- II. for the auctioneer, $p^* \in \Delta$ maximizes $p \cdot z^*$ on Δ , where $z^* = \sum_i x_i^* - \omega$

For questions (i) and (ii), if the answer is yes, prove it; if the answer is no, give a counterexample or explanation.

- (i) Is a competitive equilibrium of the exchange economy $\mathcal{E}$ an equilibrium of the abstract economy?
 - (ii) Is an equilibrium of the abstract economy a competitive equilibrium of the exchange economy $\mathcal{E}$?
- (c) Define precisely the notion of a *strategic form game*; and the notion of a *Nash equilibrium* of a strategic form game.
- (d) Is the abstract economy from (b) a strategic form game with $I + 1$ players? If so, identify each element of the game. If not, explain why it is not.